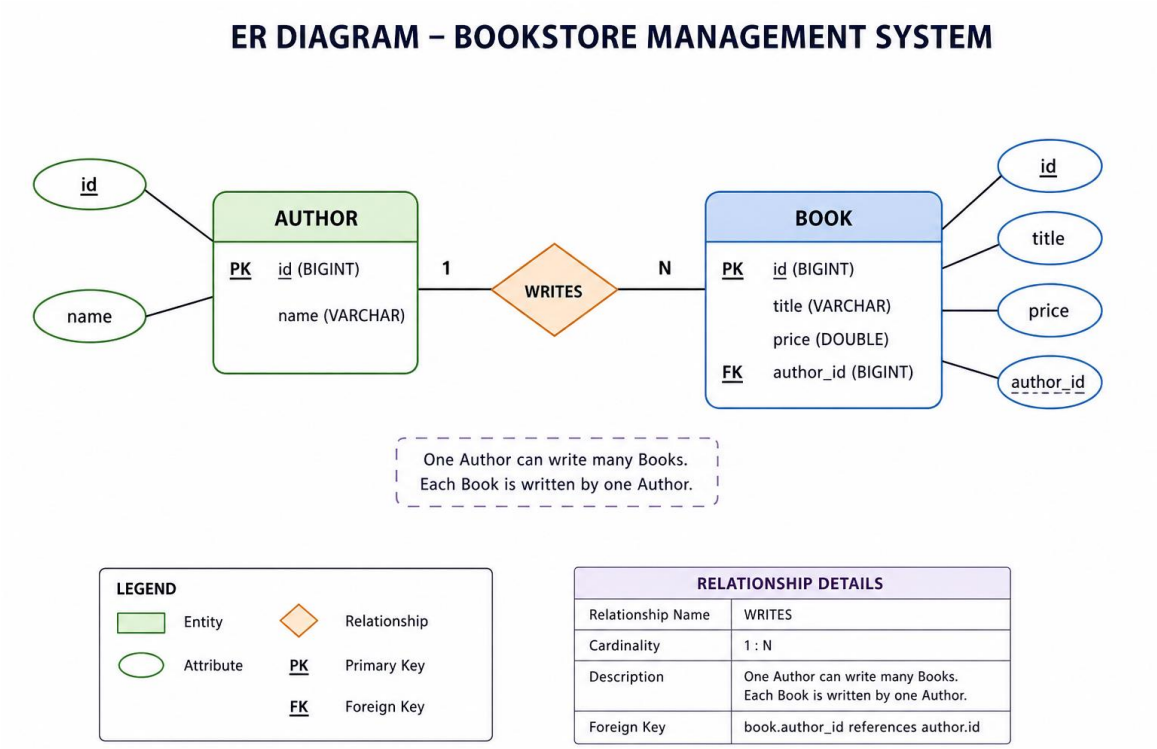
Project Title: Spring Boot Bookstore CRUD Application



BOOKSTORE PROJECT LINK

GITHUB LINK --->
https://github.com/Sanskriti10247/spring-boot-bookstore-crud.git

○ Overview

This project is a Spring Boot web application using JSP, MySQL, and JPA. It manages Books and Authors with create, read, and update operations.

– Entities

- **Author: id, name.**
- **Book: id, title, price, author (Many-to-One).**

★ Architecture

Controller -> Service -> Repository -> Database

★ Features

- ♦ Create books
- ♦ view books
- ♦ update books
- ♦ join query using JPA.

-> Database Design

Two tables: **author and book**. Foreign key relationship using *author_id.*

-> Custom Query

SELECT b FROM Book b JOIN b.author a

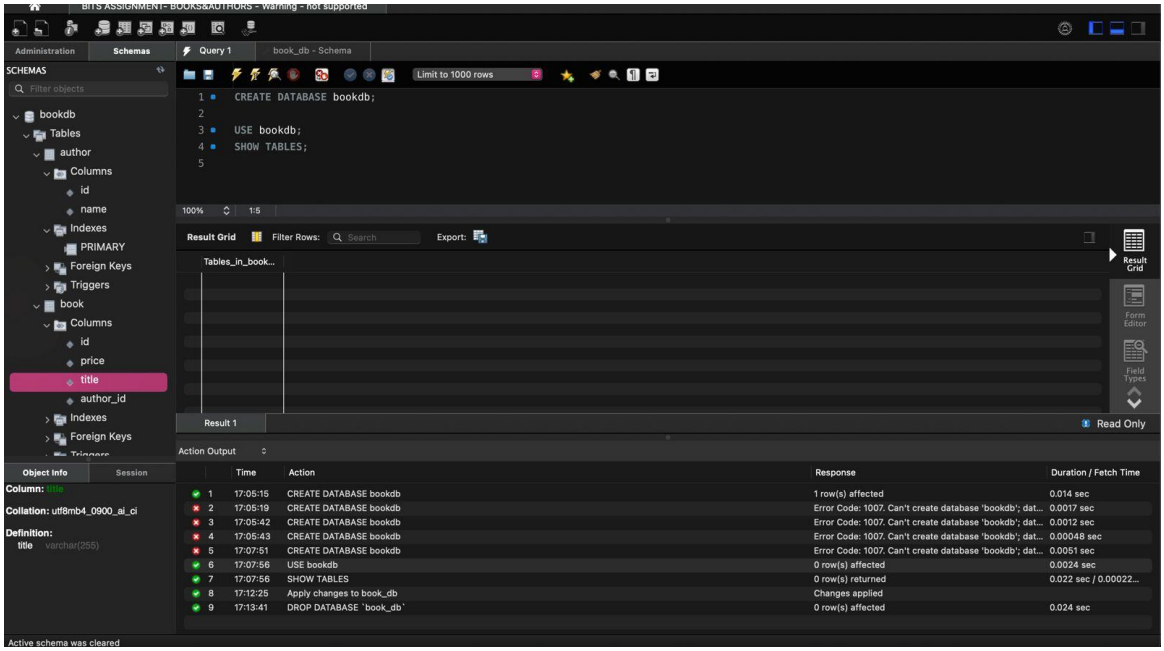
-> Challenges

Handled JSP binding, entity relationships, and edit functionality errors.

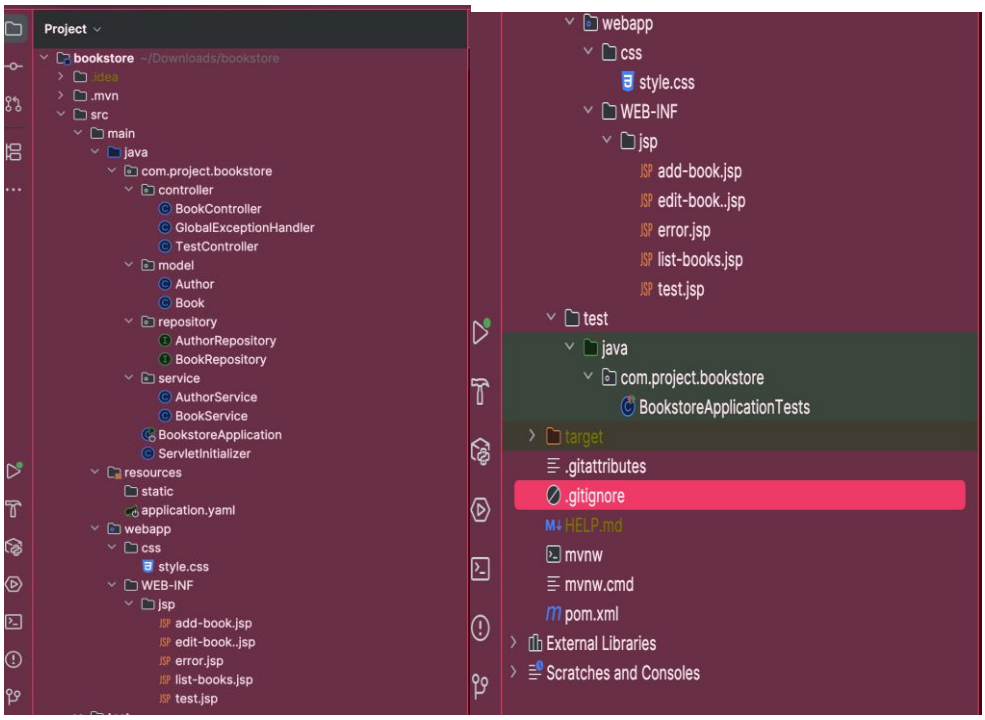
GITHUB LINK --->

https://github.com/Sanskriti10247/spring-boot-bookstore-crud.git

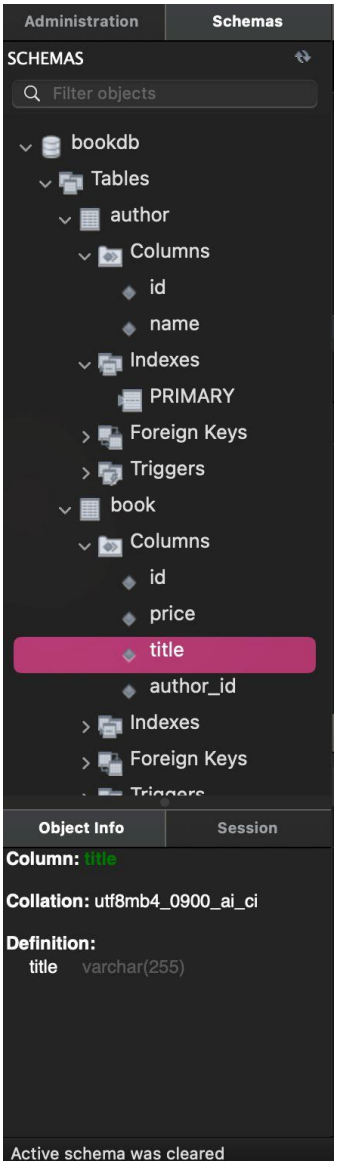
SCREENSHOTS FOR REFERENCE :-->



FOLDER STRUCTURE :

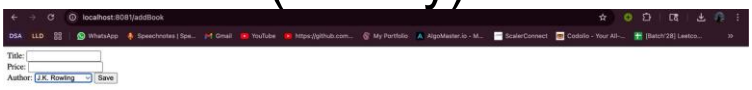


DB :



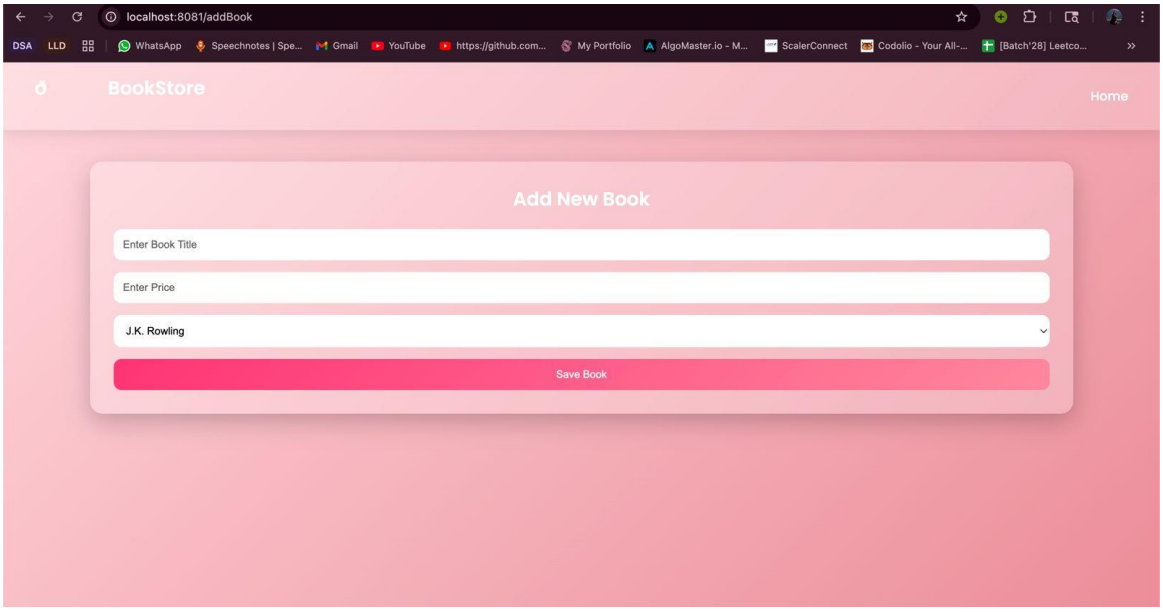
OUTPUTS->

Without UI (Initially)->

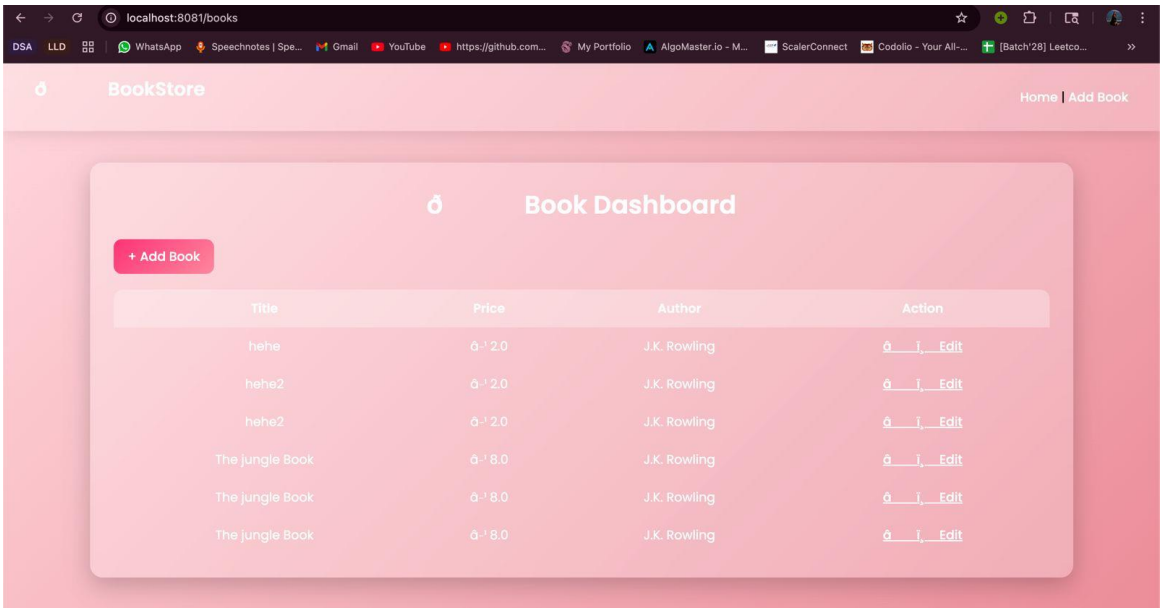


After UI Changes->

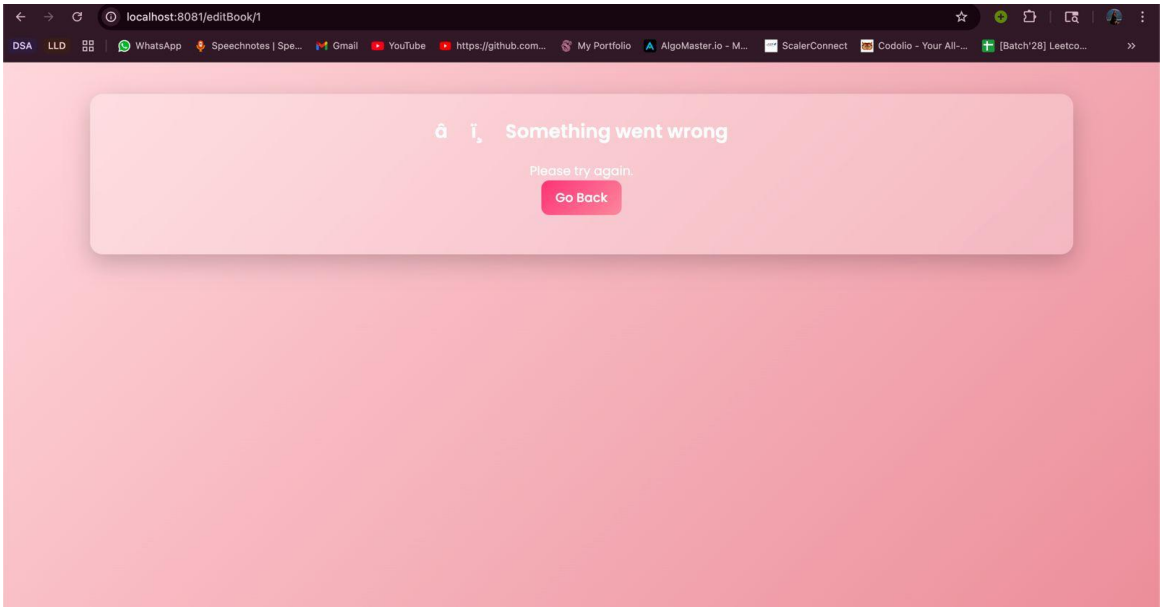
1. ADD BOOK :



2. DASHBAORD SHOWING ALL THE BOOK COLLECTIONS:



3. EDIT BOOKS :



TABLES AND THEIR RESULT IN MY SQL WORKBENCH:

SQL COMMANDS -->

1. SELECT FROM AUTHOR:

```
SELECT * FROM AUTHOR
```

2. SELECT FROM BOOKS:

```
SELECT * FROM BOOKS
```

3. (JOIN OPERATION) :

```
SELECT b.title, b.price, a.name
FROM book b
INNER JOIN author a ON b.author_id = a.id;
```

```
36  SELECT * FROM author;
```

100% 22:36

Result Grid Filter Rows: Search Edit: Export/Import:

id	name
1	J.K. Rowling
2	George Orwell
3	Agatha Christie
4	Chetan Bhagat
5	Dan Brown
6	Ruskin Bond
7	Arundhati Roy
8	Paulo Coelho
9	Jane Austen
10	Mark Twain
11	J.K. Rowling
12	George Orwell

author 6

Action Output

	Time	Action	Response
10	18:13:23	INSERT INTO author (name) VALUES ('J.K. Rowling'), ('George Orwell'), ('Chetan Bhagat'), ('Paulo Coelho'), ('Dan Brown'), ('...	10 row(s) a
11	18:13:23	INSERT INTO book (title, price, author_id) VALUES ('Harry Potter', 500, 1), ('1984', 300, 2), ('Half Girlfriend', 250, 3), ('The A...	10 row(s) a
12	18:16:37	USE bookdb	0 row(s) aff
13	18:16:37	SHOW TABLES	2 row(s) ret
14	18:16:37	INSERT INTO author (name) VALUES ('J.K. Rowling'), ('George Orwell'), ('Agatha Christie'), ('Chetan Bhagat'), ('Dan Brown'),...	10 row(s) a
15	18:16:37	INSERT INTO author (name) VALUES ('J.K. Rowling'), ('George Orwell'), ('Chetan Bhagat'), ('Paulo Coelho'), ('Dan Brown'), ('...	10 row(s) a
16	18:16:37	INSERT INTO book (title, price, author_id) VALUES ('Harry Potter', 500, 1), ('1984', 300, 2), ('Half Girlfriend', 250, 3), ('The A...	10 row(s) a
17	18:16:37	SELECT * FROM author LIMIT 0, 1000	70 row(s) r
18	18:29:20	SELECT * FROM book LIMIT 0, 1000	36 row(s) r

```
38  SELECT * FROM book;
```

100% 20:38

Result Grid Filter Rows: Search Edit: Export/Import:

id	price	title	author_id
3	2	hehe2	1
4	8	The jungle Book	1
5	8	The jungle Book	1
6	8	The jungle Book	1
7	500	Harry Potter	1
8	300	1984	2
9	250	Half Girlfriend	3
10	350	The Alchemist	4
11	450	Da Vinci Code	5
12	400	Murder on Orie...	6
13	200	Malgudi Days	7
14	300	Pride and Prej	8

book 7

Action Output

	Time	Action	Response
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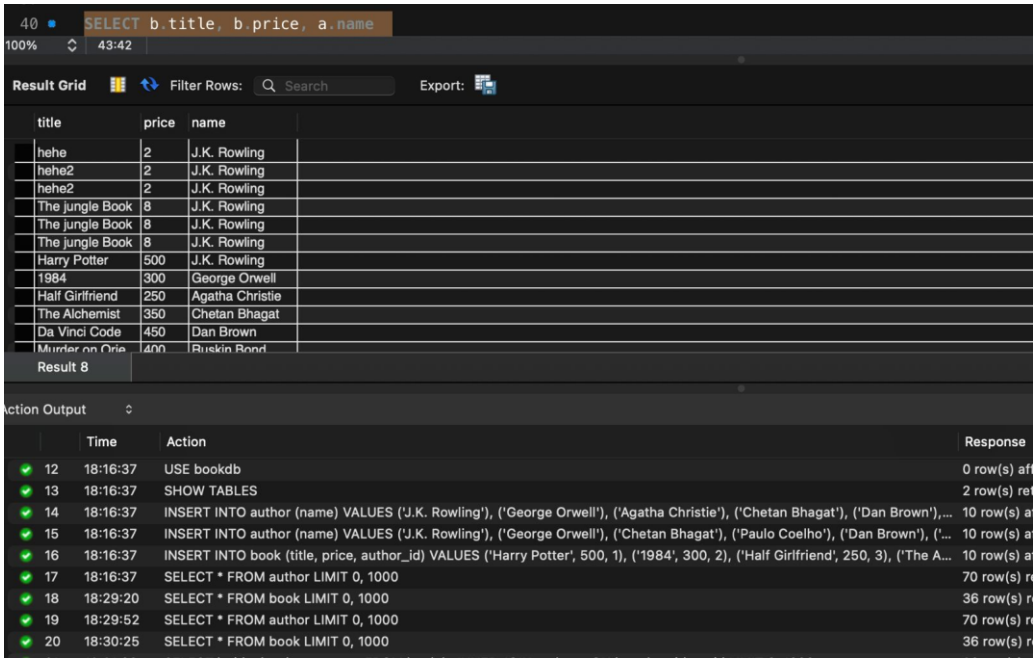
17	18:16:37	SELECT * FROM author LIMIT 0, 1000	70 row(s) r
18	18:29:20	SELECT * FROM book LIMIT 0, 1000	36 row(s) r
19	18:29:52	SELECT * FROM author LIMIT 0, 1000	70 row(s) r



```
40   SELECT b.title, b.price, a.name
```

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Result Grid Filter Rows: Search Export:

title	price	name
hehe	2	J.K. Rowling
hehe2	2	J.K. Rowling
hehe2	2	J.K. Rowling
The jungle Book	8	J.K. Rowling
The jungle Book	8	J.K. Rowling
The jungle Book	8	J.K. Rowling
Harry Potter	500	J.K. Rowling
1984	300	George Orwell
Half Girlfriend	250	Agatha Christie
The Alchemist	350	Chetan Bhagat
Da Vinci Code	450	Dan Brown
Murder on Orie	400	Ruskin Bond

Result 8

Action Output

	Time	Action	Response
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13	18:16:37	SHOW TABLES	2 row(s) ret
14	18:16:37	INSERT INTO author (name) VALUES ('J.K. Rowling'), ('George Orwell'), ('Agatha Christie'), ('Chetan Bhagat'), ('Dan Brown'),...	10 row(s) a
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18	18:29:20	SELECT * FROM book LIMIT 0, 1000	36 row(s) r
19	18:29:52	SELECT * FROM author LIMIT 0, 1000	70 row(s) r
20	18:30:25	SELECT * FROM book LIMIT 0, 1000	36 row(s) r

Conclusion

Successfully built a full-stack Spring Boot application with MVC architecture.